Introduction to Materials Property Prediction

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Part of Module 3: Approaches to Materials Property Prediction with Machine Learning

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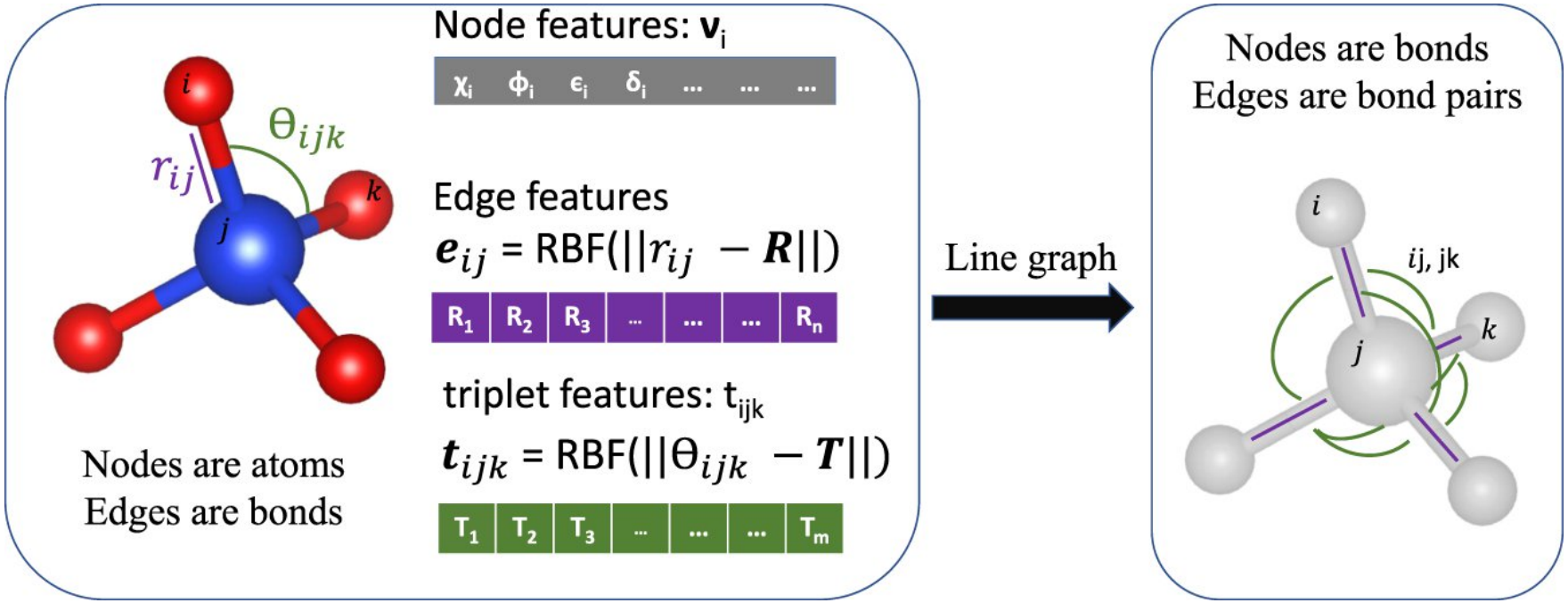
Using Atomic Structure as Features

- Mole fractions or Elemental Property features are powerful but have no structural information (e.g., will predict band gap of all polymorphs as the same).
- If you have atomic structures (very common with simulated data) it is generally better to featurize with all that information.
- A very powerful approach to this is a graph neural network.

- Example of the Atomistic Line Graph Neural Network (ALIGNN)
- Builds features from atomic structure and elemental properties.
- Has learnable updates to features followed by a simple regression on final pooled vector.



Choudhury and DeCost, npj Comp. Mat. '21

ALIGNN Tutorial Notebook

- Our ALIGNN tutorial will equip you with the tools needed to train a structure-based property predictor on your own data and for your own property of interest

- We will now move on to our lab activity, where we will:
  - Train a Voltage prediction graph neural network using UMLIP-calculated average voltages of prospective Na-cathode materials including only their structures as features.
  - Use our trained property predictor to perform a follow-up investigation of a candidate material, modifying various aspects of its structure

TMS 2025 AI Workshop: Ryan Jacobs, Ben Afflerbach, Dane Morgan (UW-Madison)

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Google Colab notebook demo of structure based property prediction:

Trainging for prediction Na-battery material voltage